

How to Download Latest Version of a File Programatically (Linux Bash)

```
LATEST_VERSION_EXT_PACK=$(wget -qO- https://www.virtualbox.org/wiki/Downloads |  
grep -o 'Oracle_VirtualBox_Extension_Pack-[0-9]\+\.[0-9]\+\.[0-9]\+' |  
sed 's/Oracle_VirtualBox_Extension_Pack-([0-9]\+)[0-9]\+.[0-9]\+/' | tr -d '[:space:]')  
  
wget https://download.virtualbox.org/virtualbox/${LATEST_VERSION_EXT_PACK}/Oracle_VirtualBox_Extension_Pack-  
${LATEST_VERSION_EXT_PACK}.vbox-extpack -P ./AUTO-INSTALLS-FILES/WORKER-TEMP
```

This Bash command is designed to extract the latest version of the Oracle VirtualBox Extension Pack from the VirtualBox Downloads webpage. Let's break it down step by step:

1. `wget -qO- https://www.virtualbox.org/wiki/Downloads`

- Purpose: Fetches the HTML content of the specified webpage.
- `wget`: A command-line utility to download content from the web.
- `-q`: Quiet mode; suppresses output other than errors.
- `-O-`: Sends the output of the downloaded content to standard output (stdout), rather than saving it to a file.

Result: The HTML source code of the VirtualBox Downloads page is passed to the next command via a pipe.

2. `grep -o 'Oracle_VirtualBox_Extension_Pack-[0-9]\+\.[0-9]\+\.[0-9]\+'`

- `grep`: A utility to search for patterns in text.
- `-o`: Outputs only the parts of the text that match the specified pattern, not the entire line.
- `'Oracle_VirtualBox_Extension_Pack-[0-9]\+\.[0-9]\+\.[0-9]\+'`:
 - Matches strings that start with `'Oracle_VirtualBox_Extension_Pack-'` followed by a version number.
 - `[0-9]` matches any digit.
 - `\+` matches one or more repetitions of the preceding digit class (escaped with `\` for compatibility with some versions of `grep`).

- ``\.`` matches a literal period.

- Example matches: ``Oracle_VirtualBox_Extension_Pack-7.0.8``,
``Oracle_VirtualBox_Extension_Pack-6.1.42``.

Result: Extracts only the strings that match the naming convention for the VirtualBox Extension Pack versions.

3. `sed 's/Oracle_VirtualBox_Extension_Pack-\(.*\)\/1/'`

- ``sed``: A stream editor for transforming text.

- ``s/.../.../``: Substitution command; replaces text that matches the first pattern with the second.

- ``Oracle_VirtualBox_Extension_Pack-\(.*\)``:

- Matches the string starting with

- ``Oracle_VirtualBox_Extension_Pack-``.

- The ``\(.*\)`` part captures the rest of the string (the version number) into a group.

- ``1``: Refers to the captured group, i.e., the version number.

Result: Strips the prefix ``Oracle_VirtualBox_Extension_Pack-`` and keeps only the version number (e.g., ``7.0.8``).

4. `tr -d '[:space:]'`

- ``tr``: A utility for translating or deleting characters.

- ``-d '[:space:]``: Deletes all whitespace characters (spaces, tabs, newlines, etc.) from the text.

Result: Removes any stray spaces or newline characters from the

extracted version number, ensuring a clean output.

Final Output

The command outputs the latest version number of the VirtualBox Extension Pack (e.g., `7.0.8`).

How It Works Together

1. Fetch HTML content of the VirtualBox Downloads page.
2. Extract versioned strings related to the Extension Pack using `grep`.
3. Clean up the string by removing the prefix and keeping only the version number using `sed`.
4. Ensure a tidy output by deleting any extra whitespace with `tr`.

This command is efficient and provides the desired information with minimal overhead.